

🇮🇹 Student Performance EDA Report

📌 1. Project Overview

This project analyzes student performance data using Exploratory Data Analysis (EDA). The goal is to identify patterns, trends, and relationships between different student attributes and academic scores.

📁 2. Dataset Description

- Dataset contains student performance records
- Includes features like math score, reading score, writing score, gender, and other attributes
- Contains 1000+ records

🧹 3. Data Cleaning Process

The following cleaning steps were performed:

- Handled missing values
- Removed duplicate records
- Corrected data types
- Checked and treated outliers
- Ensured data consistency

🇮🇹 4. Exploratory Data Analysis (EDA)

✅ Statistical Summary

Basic statistics (mean, median, std deviation) were calculated for numeric features.

✅ Distribution Analysis

Histograms were used to understand score distributions.

✅ Correlation Analysis

Relationships between variables (math, reading, writing scores) were analyzed using correlation heatmaps.

✅ Outlier Detection

Boxplots were used to detect extreme values in the dataset.

💡 5. Key Insights

- Students generally perform better in reading and writing compared to math
- Reading and writing scores have a strong positive correlation
- Few students show significantly lower performance (outliers)
- Score distribution is fairly normal with slight variations

📊 6. Conclusion

The analysis provides useful insights into student academic performance. These insights can help identify weak areas and improve educational strategies.
